

10703 Recitation 1

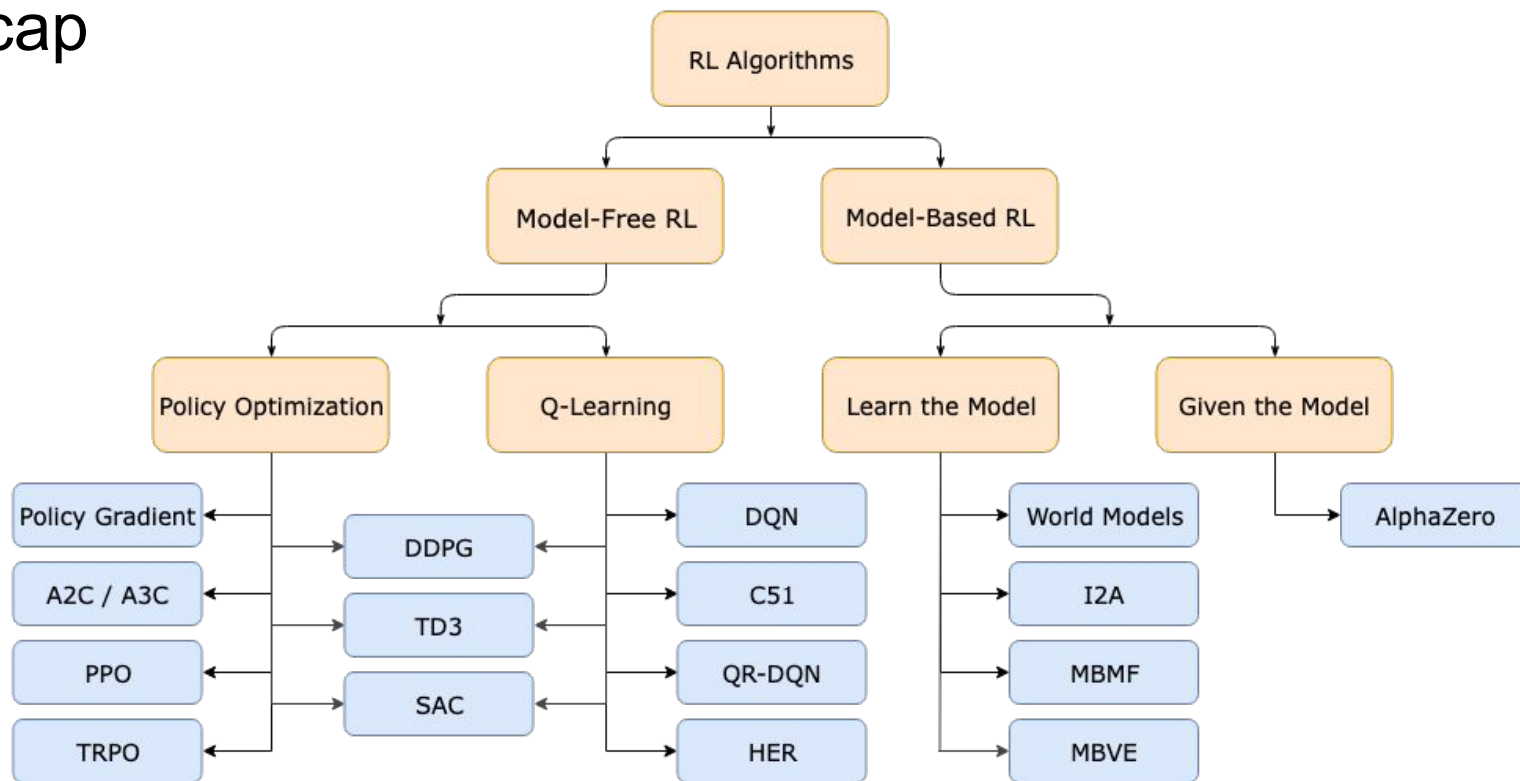
Gym and Pytorch

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Agenda

- Announcements
- Recap?
- Gym tutorial
- Pytorch tutorial

Recap



from https://spinningup.openai.com/en/latest/spinningup/rl_intro2.html

Definitions

- MDP
- Trajectories
- Return

$\mathcal{S}$ state space

$\mathcal{A}$ action space

$P(s_{t+1} \mid s_t, a_t)$ transition dynamics

$r(s_t, a_t)$ scalar reward

$\gamma \in [0, 1]$ discount factor

$P(s_0)$ initial state distribution

$$\tau = (s_0, a_0, r_0, s_1, a_1, r_1, s_2, a_2, r_2, \dots)$$

$$P_\pi(\tau) = P_\pi(s_0, a_0, s_1, a_1, \dots)$$

$$= P(s_0) \prod_{t=0}^T P(s_{t+1} \mid s_t, a_t) \pi(a_t \mid s_t)$$

$$R(\tau) = \sum_{t=0}^{\infty} \gamma^t r(s_t, a_t), \quad \gamma \in [0, 1]$$

Review: Value Functions

$$Q^{\pi}(s, a) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s, a_0 = a \right]$$

$$V^{\pi}(s) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \mid s_0 = s \right]$$

$$A_{\pi}(s, a) = q_{\pi}(s, a) - v_{\pi}(s)$$

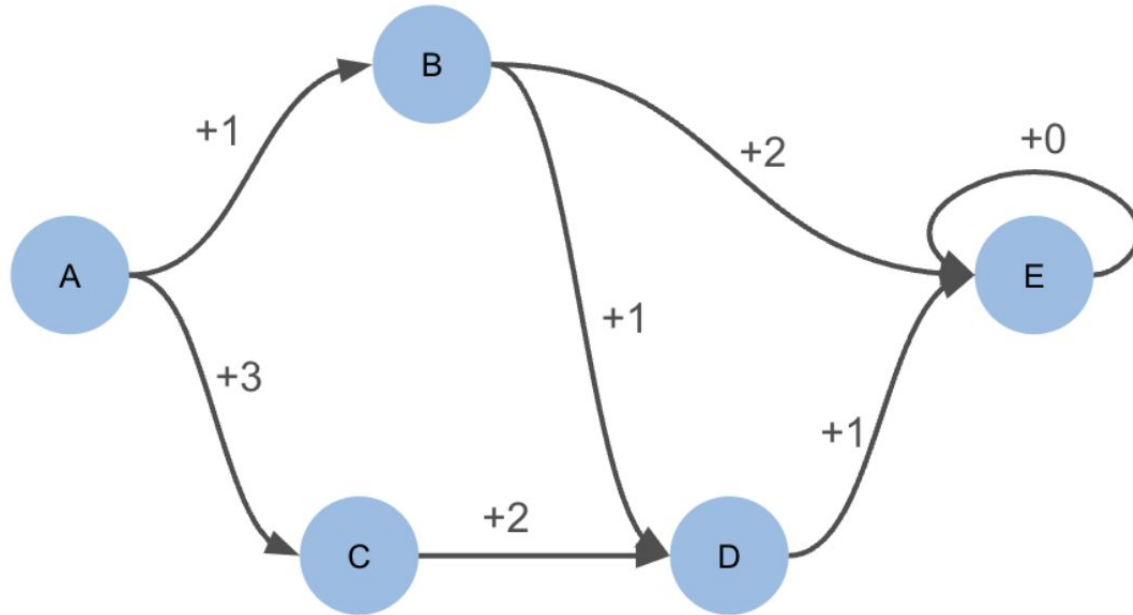
The Bellman Equations

$$Q^{\pi}(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'|s, a) \pi(a'|s')} [Q^{\pi}(s', a')]$$

$$Q^{\star}(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} \left[\max_{a'} Q^{\star}(s', a') \right]$$

$$V^{\pi}(s) = \mathbb{E}_{\pi(a|s), p(s'|s, a)} [r(s, a) + \gamma V^{\pi}(s')]$$

Policy Evaluation



Getting values with dynamic programming

Iterative Policy Evaluation, for estimating $V \approx v_\pi$

Input π , the policy to be evaluated

Algorithm parameter: a small threshold $\theta > 0$ determining accuracy of estimation

Initialize $V(s)$ arbitrarily, for $s \in \mathcal{S}$, and $V(\text{terminal})$ to 0

Loop:

$\Delta \leftarrow 0$

Loop for each $s \in \mathcal{S}$:

$v \leftarrow V(s)$

$V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) [r + \gamma V(s')]$

$\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$

But what if we don't know the dynamics???

Then we have to learn the value function (estimate it by interacting with the environment and observing rewards)

We'll start by assuming a tabular setting - small, discrete number of states and actions

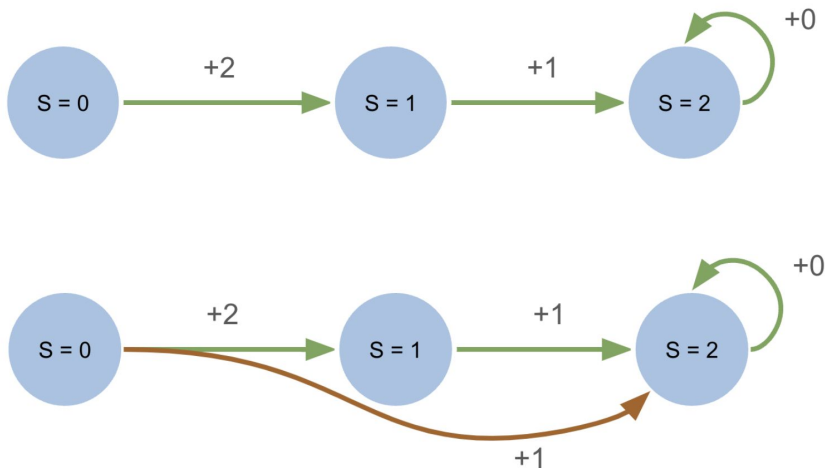
$$Q = \begin{pmatrix} \ddots & & \\ & Q(s = i, a = j) & \\ & & \ddots \end{pmatrix}.$$

Monte Carlo Estimation

Key idea: look at the state-action pairs that you have visited, and see what the future returns were afterwards.

$$V(S_t) \leftarrow V(S_t) + \alpha [G_t - V(S_t)]$$

Limitations?



TD-learning

Key idea: use predictions at one time step to update your predictions at another time step

$$V(S_t) \leftarrow V(S_t) + \alpha \left[R_{t+1} + \gamma V(S_{t+1}) - V(S_t) \right]$$

Benefits?

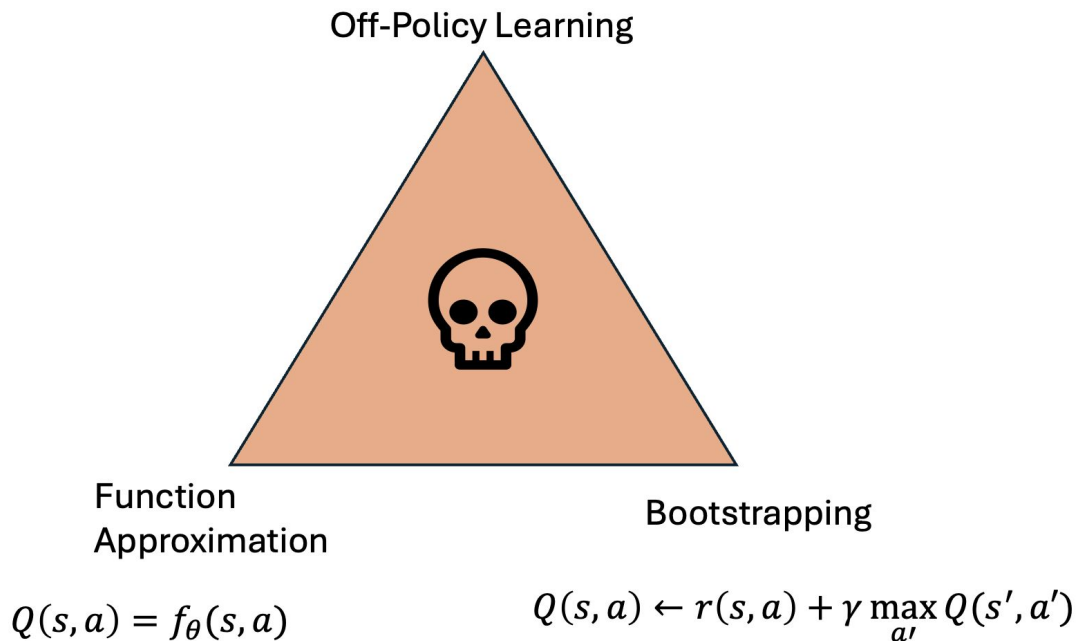
Limitations?

Imagine 10 coin flips...

MC estimate?

TD estimate?

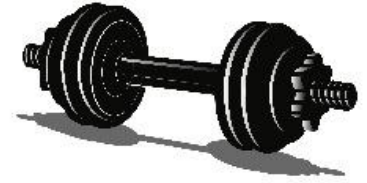
But this is really hard!



Questions?

Gym Tutorial

<https://drive.google.com/file/d/1HU2FYcLnqimdDsr9EOpSMK-HzBDJr7pw/view?usp=sharing>



 Gymnasium



Pytorch Tutorial

https://drive.google.com/file/d/15ejnTlxiRkFMs_zYMbzW1vennxfL6yEO/view?usp=sharing

References

<https://ben-eysenbach.github.io/intro-rl/>